The HexaPedal

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Problem Definition

Problem Statement

The problem of	Facilitate the usage of bike sharing in the city as well as the high emissions.
Affects	Anyone who uses bikes to commute and/or people who want low-cost transportation
The impact of which is	Promotes exercise, low-cost transportation and the promote low-emission solutions for transportation
A successful solution would be	Would be a full-fledged bike sharing system that allows the user to use all the features in the system through a clean user interface (able to reserve, ride bikes, return bikes, pay, etc.)

Product Position Statement

For	People who want to exercise, low-cost transportation, help the environment with low-emission transportation or use a bike to commute
Who	Would use HexaPedal are people trying to get from a from point A to B in in very dense cities. Also, people with more “noble” intentions such as reducing their eco-footprint.
The HexaPedal	Is a mobile application that allows you to bike share with a few-clicks of a button. Prioritising your health without comprising the usability
That	In very densely populated cities bikes are great way to get around the city all whilst saving money, exercising, and reducing your emission output
Unlike	Taking a taxi, Uber or other biking services such as Bixi’s
Our Product	HexaPedal allows you to prioritize your health anywhere. We will not be limited to one city. We will be launching docs in all major cities in Canada.

Product Overview

HexaPedal is a completely different product trying to revolutionize or improve the experience of bike-sharing in major cities. It will be accessible to all users to have access to a phone and Internet. It mainly aims to help people that want to get from point A to B that want to avoid stressing over finding parking in populated cities. This consequently allows people to reduce their eco-footprint while improving their health. It will have a very intuitive and clean user interface, that abstracts all the processing and logic going behind the scenes to give them the best and smoothest experience possible.

Assumptions and Dependencies

Assumptions	Dependencies
Our users will be using Apple Products	The entire mobile application
Use a map API (Google Maps API) [1]	The dashboard
Use a payment API (Stripe API) [2]	Checkout page/dashboard
Using a third-Party email service (Resend or other)	Backend

Technology used

Control version system

We will be using the entire GitHub ecosystem. To assign tasks, raise bugs, etc. We will use GitHub Issues. To visualize the tasks for a specific period we will use GitHub Projects. In need of documentation notes we be taken in GitHub wiki. Finally, to ensure any new changes do not break production we will using GitHub Actions to implement our CI/CD pipeline.

Team collaboration

We will be following Agile practices and set Scrum meetings on a weekly or bi-weekly basis to ensure the smooth progress of the project. Also, to give updates on our specific assigned tasks to make sure there are no blockers and no on needs help. This ensures consistent and constant communication between all team members.

Design and modelling work

All modelling and design work be on either Draw.io or Visual Paradigm. For example, the context diagram is being built on Draw.io whilst the Domain model for the application is being designed on Visual Paradigm.

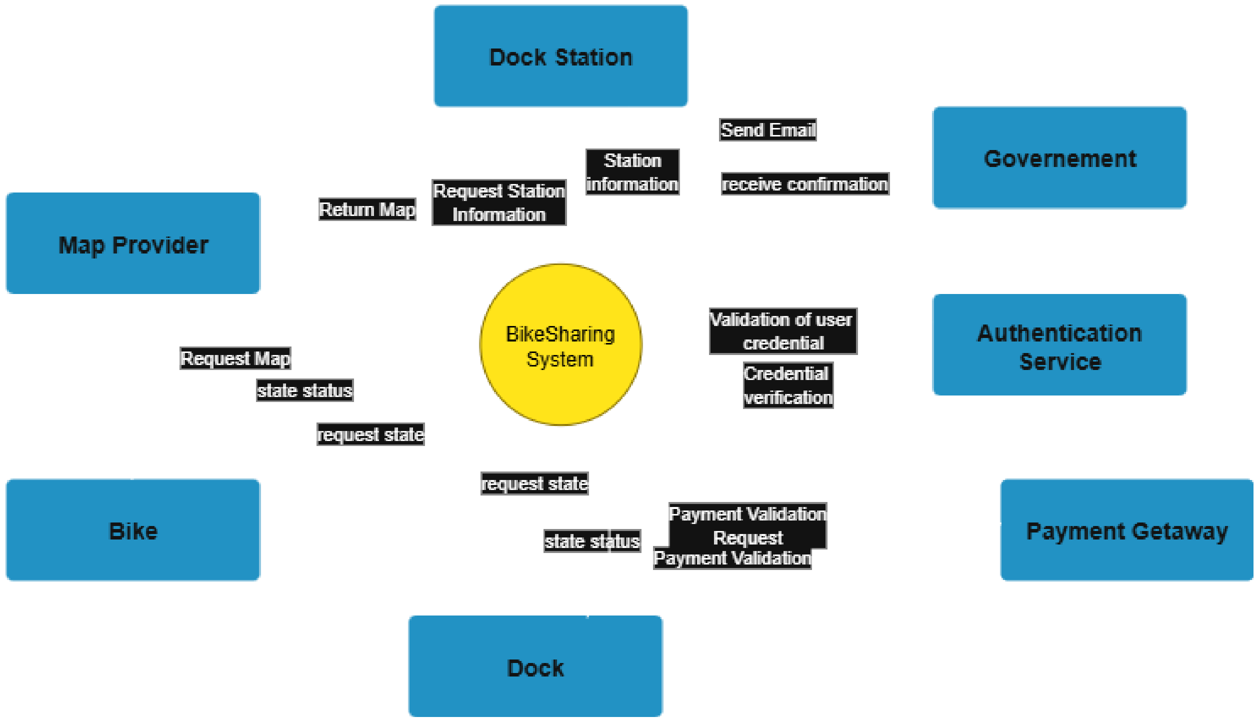
Development framework

For the frontend we will be using Next.js, a component library, and Tailwind CSS for styling. For the backend, we will be using Spring Boot, and Supabase for the database and authentication using OAuth. For external third-party APIs, we will utilize Google Maps API for the map, Stripe API for the entire payment flow, and either the Gmail SMTP server or third-party service like Sendgrid, Resend, etc.

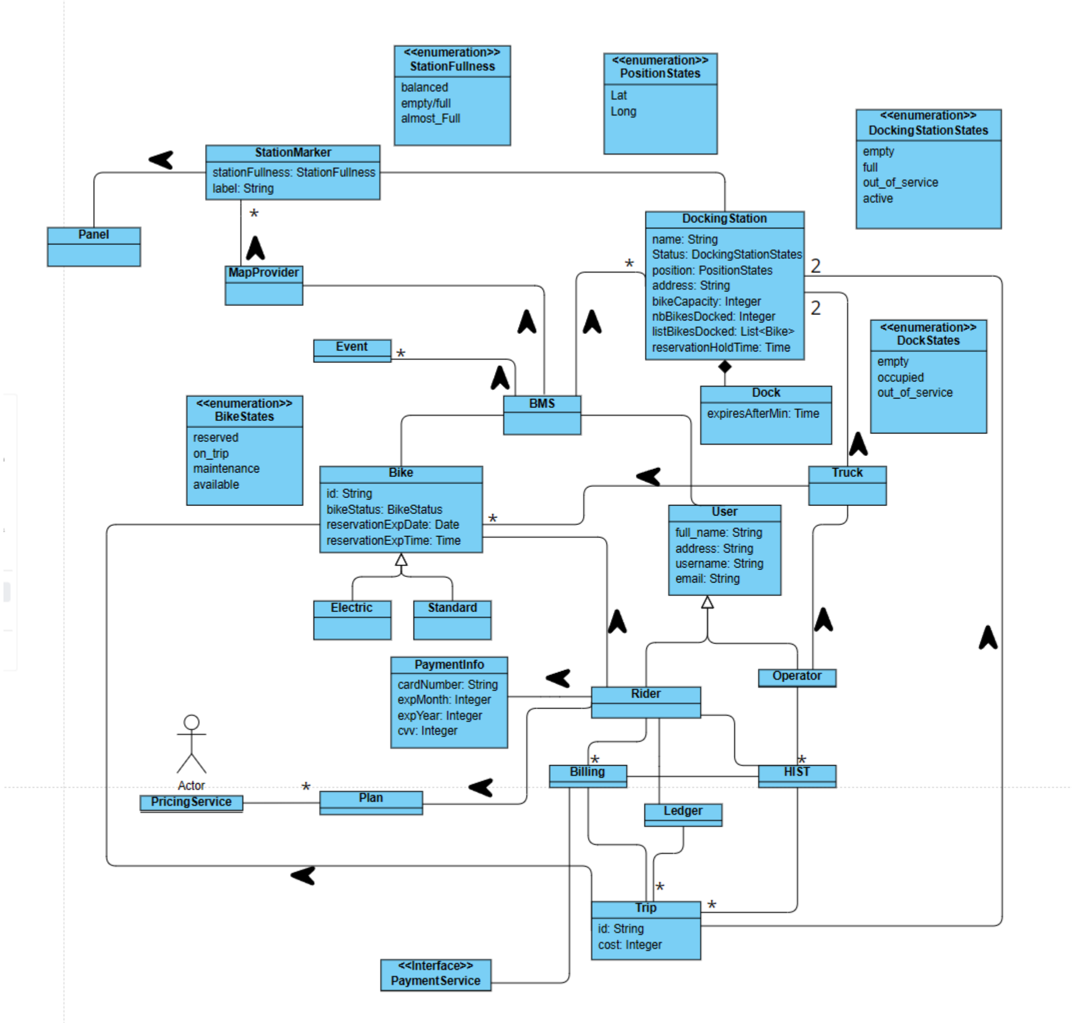
Coding

The programming languages being used are TypeScript, Java, and SQL (to create the tables, indexes and RLS).

Context Diagram



Domain Model



References

[1] https://developers.google.com/maps

[2] https://docs.stripe.com/api